

HLD Chapter 2: Back-of-the-Envelope Estimations Masterclass

Core Learning Objective: Master the mathematical formulas, rules of thumb, and conversion shortcuts required to estimate QPS, storage, bandwidth, and cache sizes in under 5 minutes during a systems design interview.

1. Golden Rules of Thumb & Approximation Constants

In an interview, **never spend time doing manual long division with decimals**. Interviewers expect clean, fast order-of-magnitude estimations (10^x).

The 86,400 Approximation Rule

$$1 \text{ day} = 24 \text{ hours} \times 60 \text{ minutes} \times 60 \text{ seconds} = 86,400 \text{ seconds} \approx 10^5 \text{ seconds}$$

$$\text{QPS} = \frac{\text{Daily Requests}}{10^5}$$

- 10 Million requests/day $\approx \frac{10^7}{10^5} = 100$ QPS
- 100 Million requests/day $\approx \frac{10^8}{10^5} = 1,000$ QPS
- 1 Billion requests/day $\approx \frac{10^9}{10^5} = 10,000$ QPS

Peak Traffic Multiplier

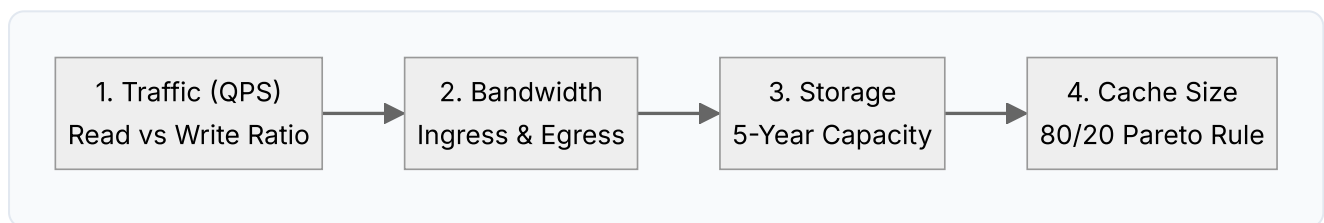
Systems must be provisioned for **Peak QPS**, not average QPS:

$$\text{Peak QPS} = \text{Average QPS} \times 2 \text{ (or } 3 \times \text{ for spiky apps like Twitter/Uber)}$$

2. Storage & Memory Units Mapping

Power of 2	Power of 10	Value	Unit	Concrete Example
2^{10}	10^3	1,024	Kilobyte (KB)	Small JSON payload / metadata
2^{20}	10^6	1,048,576	Megabyte (MB)	High-res compressed photo
2^{30}	10^9	1,073,741,824	Gigabyte (GB)	Modern RAM memory stick
2^{40}	10^{12}	10^{12}	Terabyte (TB)	Hard drive disk capacity
2^{50}	10^{15}	10^{15}	Petabyte (PB)	Datacenter storage tier

3. The 4-Step Estimation Blueprint



Step 1: Traffic Estimation (Read vs Write QPS)

1. Identify **Daily Active Users (DAU)**.
2. Determine user actions per day (e.g. 10 reads, 1 write per user).
3. Compute Read QPS and Write QPS.

Step 2: Storage Requirements (5-Year Capacity)

$$\text{Daily Storage} = \text{Writes/Day} \times \text{Size per Record}$$

$$\text{5-Year Storage} = \text{Daily Storage} \times 365 \times 5 \approx \text{Daily Storage} \times \mathbf{2,000}$$

Step 3: Network Bandwidth (Ingress & Egress)

- **Ingress (Incoming):** Write QPS $\times$ Average Write Payload Size
- **Egress (Outgoing):** Read QPS $\times$ Average Read Payload Size

Step 4: Cache Memory Sizing (The 80/20 Pareto Principle)

In almost every consumer system, **20% of daily read requests account for 80% of total read volume** (the "hot" data).

$$\text{Cache RAM} = \text{Daily Total Read Data Volume} \times \mathbf{0.20}$$

4. End-to-End Walkthrough: Twitter-Scale Social Platform

Assumptions & Given Metrics:

- **DAU:** 300 Million active users.
- **Writes:** Each user posts an average of 2 tweets per day.
- **Reads:** Each user views an average of 20 tweets per day.
- **Media:** 10% of tweets contain an image (200 KB), 90% are pure text (200 bytes).

1. QPS Calculation:

- **Total Writes/Day:** $300\text{M} \times 2 = 600 \text{ Million tweets/day}$.

$$\text{Write QPS} = \frac{600 \times 10^6}{10^5} = \mathbf{6,000 \text{ QPS}} \quad (\text{Peak} = 6,000 \times 2 = \mathbf{12,000 \text{ QPS}})$$

- **Total Reads/Day:** $300\text{M} \times 20 = 6 \text{ Billion reads/day}$.

$$\text{Read QPS} = \frac{6 \times 10^9}{10^5} = \mathbf{60,000 \text{ QPS}} \quad (\text{Peak} = 60,000 \times 2 = \mathbf{120,000 \text{ QPS}})$$

- **Read-to-Write Ratio:** 10 : 1 (Read-heavy architecture).

2. Storage Calculation:

- **Average Tweet Size:** $(0.90 \times 200 \text{ B}) + (0.10 \times 200 \text{ KB}) \approx 180 \text{ B} + 20 \text{ KB} \approx \mathbf{20 \text{ KB}}$.
- **Daily Storage:** $600\text{M writes} \times 20 \text{ KB} = 12 \text{ TB / day}$.
- **5-Year Storage:** $12 \text{ TB} \times 365 \times 5 \approx \mathbf{22 \text{ PB}}$ (Petabytes).

3. Cache Sizing (80/20 Rule):

- **Daily Read Volume:** $6 \text{ Billion reads} \times 20 \text{ KB} = 120 \text{ TB / day}$.
- **Cache 20% of Hot Daily Data:** $120 \text{ TB} \times 0.20 = \mathbf{24 \text{ TB}}$ of RAM.
- If each Redis cache server has 256 GB of RAM:

$$\text{Number of Redis Nodes} = \frac{24 \text{ TB}}{0.256 \text{ TB}} \approx \mathbf{94 \text{ Cache Instances}}$$

5. Latency Numbers and a Reusable Calculator

Orders of magnitude worth memorising

Operation	Rough time	Use it to decide
Main memory reference	100 ns	An in-process cache is effectively free
Read 1 MB sequentially from memory	10 to 50 microseconds	Why you cache hot data in RAM
SSD random read	100 microseconds	Databases on SSD handle thousands of reads per second per disk
Round trip inside one data centre	0.5 ms	A chain of ten service calls costs about 5 ms of pure network
Read 1 MB sequentially from SSD	about 1 ms	Large scans are bandwidth-bound
Disk seek (spinning disk)	5 to 10 ms	Why random access on HDD is slow
Cross-continent round trip	100 to 150 ms	Why you put data near users (CDNs, regional replicas)

These are order-of-magnitude figures; hardware changes them by factors of two or three, but the ratios (memory about 1,000 times faster than SSD, SSD about 100 times faster than a cross-continent call) stay stable.

One function for every estimation question

How to use the three results: URL shortener storage is **single-digit terabytes** (a database), chat text is **a few petabytes a year** (a sharded wide-column store), and photos are **hundreds of petabytes** (an object store plus a CDN). The estimate tells you which architecture family you are in before you draw anything.

Mistakes to avoid

1. Forgetting the peak factor: average QPS sizes the budget, peak QPS sizes the servers.
2. Forgetting replication in storage (a factor of 3 is easy to omit).
3. Mixing bits and bytes in bandwidth (a 1 Gbps link carries about 125 MB per second).
4. Quoting false precision: say "about 10 TB", not "9.13 TB"; the inputs are guesses.

5. Skipping the sanity check: does one server's worth of memory, disk or network hold this? If yes, do not shard.
-

Further Reading

- [Latency numbers every programmer should know](#)
 - [System Design Primer: appendix](#)
 - [AWS Builders' Library](#)
-

Check Yourself

Answer in your head or on paper first, then open each answer.

- ▶ 1. Approximate seconds in a day, for QPS maths?
- ▶ 2. 1M requests per day is roughly how many QPS?
- ▶ 3. How much storage is 1 KB per record for 1 billion records?
- ▶ 4. Why show your arithmetic in an interview?